

Digital Rehabilitation Research Suite

Technical Project Description, Algorithmic Specification & Validation Roadmap

Physical Limitations Index (PLI) · Cervical ROMLens · BreathScope · GaitScope / Biomechanics Engine · EncounterGuard v8

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1. Executive Summary

This document provides a complete, code-accurate technical description of a suite of browser-based research prototypes developed for physiotherapy and psychosomatic rehabilitation. All tools run entirely client-side (no cloud dependency for core measurement), emphasise privacy-by-design, and are explicitly positioned as non-medical-device research instruments under an L1 validity / EU MDR framework-awareness approach.

The suite consists of five integrated components:

- Physical Limitations Index (PLI) – modified Neck Disability Index with FHIR, HL7 v2, PHR and SMART-goal generation
- Cervical ROMLens PRO – MediaPipe Pose-based selfie-camera cervical range-of-motion measurement with FHIR export
- BreathScope – microphone-based relative respiratory screening (RMS peak → estimated PEF/FEV₁)
- GaitScope / Biomechanics Engine (V27) – MediaPipe Pose gait and joint-angle analysis
- EncounterGuard v8 – lean physiotherapy encounter CDSS with ICD lookup, ICF-informed functional screen, safety flags and transparent risk heuristic

2. Regulatory & Scientific Positioning

All tools are research prototypes. They are not CE-marked medical devices under the EU Medical Device Regulation (MDR). They are intended solely for research, screening, trend monitoring, workflow exploration and institutional evaluation. Clinical decisions must remain the responsibility of qualified practitioners.

The developer maintains an explicit L1 Validity framework and EU MDR awareness. Participant information and study protocols must state the investigational status of the digital instruments.

3. Tool-by-Tool Technical Specification (Code-Accurate)

3.1 BreathScope – Microphone-Based Respiratory Screening

Core Algorithm

BreathScope uses the Web Audio API and getUserMedia to capture a 5-second forced exhalation via the device microphone.

- Audio is analysed in the time domain with AnalyserNode (fftSize = 2048).
- For each animation frame the RMS (root-mean-square) amplitude of the normalised time-domain buffer is computed.
- The maximum RMS value recorded during the 5-second window is stored as peak.
- Relative PEF is derived as: $\text{peak} \times 800$.
- Estimated FEV₁ (litres) is calculated as: $\text{PEF} \times 0.004 \times (\text{height}/170) \times (40/\text{age})$.
- A simple expected FEV₁ formula is applied: $0.04 \times \text{height} - 0.03 \times \text{age} - 2$.

- Percentage of expected is clamped between 0 % and 100 % and colour-coded (red < 50 %, amber 50–80 %, green ≥ 80 %).

An optional Clinical Validation Mode allows the user to enter a reference spirometer FEV₁ value for percentage-difference comparison. Trend data (up to 50 points) are stored in localStorage and visualised as a simple line chart.

Disclaimer rendered in the UI: “Not a medical device. Screening and trend monitoring only.”

3.2 Cervical ROMLens PRO – Selfie-Camera Cervical ROM

Core Algorithm

ROMLens forces the front/selfie camera and uses MediaPipe Pose (modelComplexity 1, minDetectionConfidence 0.6, minTrackingConfidence 0.5).

- User sets a personal neutral (zero) position for the selected plane.
- Four primary planes are supported: Flexion, Extension, Rotation, Lateral Flexion.
- Landmark-derived angles are tracked in real time; maximum absolute deviation from neutral is retained per plane.
- Clinical interpretation compares measured maxima against simple reference ranges and assigns Normal / Reduced / Severe status.
- FHIR R4 Observation Bundle export is generated with LOINC-inspired codes for each plane, valueQuantity in degrees, and device extension identifying Cervical ROMLens.

Session history and movement time-series are persisted in localStorage. Sharing options include WhatsApp text, e-mail, JSON and FHIR.

3.3 GaitScope / Biomechanics Engine (V27)

Core Algorithm

The engine uses MediaPipe Pose on the environment-facing camera.

- Knee angle is computed from landmarks hip–knee–ankle via vector angle formula.
- Hip angle is computed from shoulder–hip–knee.
- Step detection triggers when knee angle falls within a flexion window and a minimum inter-step interval (300 ms) is respected.
- Cadence (steps/min) is derived from recent step timestamps.
- Stride length is estimated from left–right ankle horizontal distance scaled by a calibration factor.
- Asymmetry is calculated from left–right hip vertical difference.
- A simple composite risk score combines deviation of knee angle from 170° and asymmetry.

Real-time metrics, a live Chart.js plot, JSON/CSV/PDF/FHIR export and a printable clinical report are provided.

3.4 Physical Limitations Index (PLI)

Core Logic

PLI is a 10-item patient-reported questionnaire (0–5 ordinal scale per item, total 0–50). Percentage disability and five severity bands (Minimal → Very Severe) are calculated. Each item is mapped to an ICF code.

- Dynamic evidence-based intervention suggestions and a one-click SMART goal generator are included.
- Full FHIR R4 support: Questionnaire, QuestionnaireResponse, Observation (total and percent), Condition and ServiceRequest resources packaged as a Bundle.

- HL7 v2 ORU^R01 message generation with OBX segments for each item plus total/percent/level.
- Personal Health Record (PHR) JSON export and printable clinical report.

The instrument is a deliberate modification of the classic Neck Disability Index, tailored for broader physical-limitations indexing in psychosomatic and physiotherapy settings.

3.5 EncounterGuard v8 – Physiotherapy Encounter CDSS

Core Logic

EncounterGuard is a lean, human-in-the-loop clinical decision support prototype designed for physiotherapy encounters.

- Manual ICD-10 code entry with type-ahead search across a library of 300+ codes; primary/secondary designation.
- Four ICF-informed functional domains scored 0–10 by single-click buttons: Pain, Mobility, Fall risk, Fatigue.
- One-click safety / red-flag screen (Chest pain, Neuro change, Respiratory, Trauma, Systemic).
- Transparent heuristic risk engine: ICF sum + clinical-load weighting derived from selected ICD chapters (higher weights for acute cardiac/neurological codes).
- Output status: Green / Yellow / Red together with explicit rationale and recommendations.
- Exports: PDF report, FHIR Bundle (Patient + Conditions + Observations), QR-code summary, e-mail draft.

The tool explicitly states that it is a prototype heuristic, not a validated risk score, and that human clinical judgement is required.

4. Multimodal Validation Protocol (Research Use)

Because none of the digital measurement tools are CE-marked medical devices, formal clinimetric evaluation is required before any research claims can be made. The following protocol maximises efficient use of local rehabilitation resources by assessing all tools on the same participant cohort.

4.1 Study Design

Cross-sectional clinimetric study with repeated measurements (test–retest interval 24–72 h). Primary focus: reliability and concurrent validity where a reference standard exists; feasibility and association analyses for all tools.

4.2 Participants

Adults (18–65 years) with neck-related or gait-related physical limitations attending psychosomatic or musculoskeletal rehabilitation. Target sample: 40–60 participants with complete data. Ethics approval and explicit investigational-device wording in the participant information sheet are mandatory.

4.3 Reference Standards & Procedures

- Cervical ROM: universal goniometer or validated digital inclinometer versus ROMLens (randomised order).
- BreathScope: optional concurrent comparison with a clinical spirometer FEV₁ when available; otherwise test–retest reliability and association with PLI / functional measures.
- GaitScope: concurrent comparison of selected metrics (cadence, step count) against visual count or validated wearable where feasible.

- PLI: internal consistency, test–retest, and convergent validity against established disability scales.
- EncounterGuard: inter-rater reliability of the structured data entry and descriptive evaluation of the heuristic output against clinician global impression.

4.4 Statistical Analysis

ICCs (with 95 % CIs), SEM and MDC for continuous measures; weighted kappa for ordinal checklist items; Pearson/Spearman correlations and Bland–Altman plots for concurrent validity; descriptive feasibility metrics (completion time, technical success rate, usability ratings).

5. Integration Architecture

All tools share a common design philosophy:

- Pure client-side execution (Web Audio, MediaPipe, localStorage)
- Explicit non-medical-device labelling
- FHIR R4 export pathways (Observation, QuestionnaireResponse, Bundle)
- Privacy-by-design / cloud-free measurement core
- Bilingual (EN/DE) user interfaces
- L1 Validity framing and EU MDR awareness statements

6. Recommendations for PhD Thesis Integration

- Treat the suite as a coherent digital health contribution: instrument development (PLI) + objective measurement tools + encounter-level CDSS.
- Dedicate a methods chapter (or discrete papers) to the clinimetric evaluation protocol outlined above.
- Publish algorithmic descriptions exactly as implemented (RMS peak for BreathScope, MediaPipe landmark angles for ROMLens/GaitScope, heuristic weighting for EncounterGuard).
- Maintain strict separation between research findings and any future regulatory pathway.
- Deposit frozen versions of the HTML/JS source and a software management plan as thesis supplements.

7. Contact & Document Control

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